

CERN Summer Internship | June 2026

Media Coverage

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LEADERS OF CHANGE



LEARNING PARTICLE PHYSICS AT CERN

A team of bright young Indian students selected from a cohort of international students have been successfully working on a key project at CERN (Geneva, Switzerland), the globally renowned European Organization for Nuclear Research, to study the design, operation and educational applications of the Cloud Chamber, one of the earliest particle detectors capable of making otherwise invisible cosmic particles visible.

The six-member Indian team — Ved Rathi, Arya Rathi, Suchir Divekar, Samyak Mahanty, Pragya Goyal and Sushrut Karmarkar — have been working alongside senior scientists at the world's leading particle physics laboratory. Scientists at CERN have been actively mentoring this team and promoting advanced scientific learning through international research exposure. The initiative aims to bridge the gap between cutting-edge scientific research and classroom education by introducing practical, inquiry-based learning experiences for school students.

EXPLORING THE SCIENCE BEHIND THE CLOUD CHAMBER

During their internship, the students explored the scientific and engineering principles behind the cloud chamber, including thermal management, insulation, particle detection and experimental design. Their work combined concepts of physics, engineering and scientific communication while examining how such technology can be adapted for educational use in schools. The project envisioned by the Lifelong Foundation aims to provide students with hands-on experimental experiences that stimulate curiosity and encourage scientific thinking from an early age instead of limiting the study of atoms, radiation and particle physics only from textbooks.



Taking Indian science learning to CERN

A team of six Indian students has taken classroom learning to CERN in Geneva, exploring Cloud Chamber technology and gaining hands-on exposure to particle physics, engineering and scientific research

MENTORSHIP AT THE WORLD'S LEADING PARTICLE PHYSICS LABORATORY

Commending the students' performance, Pieter Everaerts, Senior

Physicist with the CMS Experiment at CERN said, "We highly value the curiosity and professionalism of the young Indian students whom the foundation brings to CERN. They have consistently impressed us with their scientific competence, adaptability, and collaborative spirit. It has been a pleasure to mentor such motivated students, whose dedication and strong work ethic speak volumes about their potential. Ultimately, scientific talent thrives wherever young people are provided with opportunity, trust, and supportive mentorship."

The students experienced CERN's collaborative research environment, interacted with leading physicists and engineers, and gained first-hand exposure to the



The teams inspiration...CERN's Globe of Science and Innovation!

sophisticated technologies used in experiments at the world famous Large Hadron Collider (LHC) and the Globe of Science and Innovation. The experience also highlighted the importance of teamwork, precision, perseverance and problem-solving in scientific research.

for guiding us throughout the project. Overall, it was an unforgettable experience that broadened both our knowledge and our perspective."

A TRANSFORMATIVE EXPERIENCE

Reflecting on the internship, team member Ved Rathi said, "Our internship at CERN is a remarkable experience through which we gained valuable exposure and skills while learning how to solve complex scientific challenges. Exploring the advanced physics labs and equipment strengthened our passion for STEM education. Furthermore, the insightful lectures and the intellectual discussions with senior leadership at CERN further inspired us to pursue research-driven innovation in the future. We would like to thank our mentors at CERN



FROM STUDENTS TO SCIENCE AMBASSADORS

There is another important dimension to the students' journey. They should not be seen solely as beneficiaries of education; they can also become its designers, mentors and ambassadors.

"Having studied the Cloud Chamber through direct experience, the students can now help future cohorts understand both the science and the process of experimentation. They can share what worked, what needed to be changed and what they learned from difficulties encountered along the way," says Dr Archana Sharma, Senior Scientist at CERN.

BRINGING CERN'S LEARNING BACK TO INDIAN CLASSROOMS

One of the project's long-term objectives is to establish cloud chamber-based educational programmes in schools across India, supported by trained teachers, classroom resources and practical science modules. This initiative is expected to encourage greater interest in STEM education, inspire future scientists and innovators, and provide young learners an opportunity to witness the invisible world of particle physics through real scientific experimentation.

By connecting Indian classrooms with CERN, one of the world's most advanced scientific institutions, the initiative represents an important step towards nurturing scientific curiosity, strengthening research culture and preparing the next generation of innovators to contribute to global scientific advancement.

CMS Upgrade: Detector Laboratory Work

During my summer internship at CERN, I participated in laboratory activities associated with the CMS GEM detector upgrade. My work included assembly, sensor mounting and detector preparation. Handling detector components in a research environment gave me practical exposure to the precision, care and safety required when working on scientific instrumentation.





Cloud Chamber Kit: Making Particle Physics Accessible

I worked with a team to make particle physics more accessible to Indian school students through an affordable cloud chamber kit. Conventional cloud chambers use dry ice to reveal the tracks of charged particles, but sourcing, handling and cost can make them difficult for schools to use.

I investigated alternative cooling methods and took the lead in designing the kit's packaging. I explored a gel-pack-based cooling system as a possible alternative to dry ice. However, testing revealed that a conventional refrigerator could not achieve the required temperature, while keeping the gel pack sufficiently cold for sustained use was impractical. This helped me understand that a promising design must work under real-world constraints, not just in theory.

I also evaluated locally available components for a kit estimated to cost approximately ₹3,000 (USD 30). In designing its packaging, I focused on component organisation, safety guidance and ease of use for students and educators. I further helped students at BBLC construct a working cloud chamber, bringing our team's work into a hands-on learning environment.

